Structures 69 (2024) 107425



Contents lists available at ScienceDirect

Structures

journal homepage: www.elsevier.com/locate/structures





Optimization method of cable structure demolition driven by digital twin evolution model

Guoliang Shi [a,b], Zhansheng Liu [a,b,*], Dechun Lu [a,b], Qingwen Zhang [a,b]

[a] Faculty of Architecture, Civil and Transportation Engineering, Beijing University of Technology, Beijing 100124, China
[b] The Key Laboratory of Urban Security and Disaster Engineering of the Ministry of Education, Beijing University of Technology, Beijing 100124, China

ARTICLE INFO

Keywords:
Cable structure
Digital twin
Structure demolition
Optimization of scheme
Experimental verification

ABSTRACT

The large-span cable structure has the problems of complex component connection, cumbersome demolition process and difficult process coordination, which lead to high construction safety risk and low efficiency. Moreover, the optimization of the construction process scheme is difficult, and the convergence and visualization of the finite element analysis are insufficient, which hinders the construction efficiency. This study proposes a cable structure demolition optimization method driven by the digital twin (DT) evolution model. From the perspective of virtual-real mapping, the DT evolution process of cable structure demolition is clarified. The virtualization method of physical demolition process is formed by integrating visual modeling, simulation modeling and data modeling. Taking full account of the temporal and spatial evolution of the demolition process, a DT evolution model of the demolition process is established. Starting from the information flow, component flow, control flow and their relationship, the problems existing in the process of demolition execution are analyzed. Driven by the twin evolution process, a data mapping method for the demolition process is proposed. The safety risk is characterized by the mechanical properties (MPs) of the structure, and the expression language of the data association relationship is established. The particle swarm optimization back propagation neural network (PSO-BPNN) is improved, and an intelligent analysis method of MPs in the whole process of demolition is proposed. The best construction scheme is obtained by integrating the changes in MPs between the various demolition steps. The information of the demolition evolution process is divided into functional modules, and a DT platform is developed to efficiently guide the actual construction. Taking the demolition process of a cable truss experimental model as an example, the effectiveness of the proposed method is verified. The research results show that the DT evolution model can accurately map the actual demolition process. The deep learning method analyzes the mechanical properties of the structure and obtains the best construction scheme quickly and accurately. The development and application of the twin platform improves the degree of visualization and realizes the efficient guidance of the virtual model to the actual construction.

1. Introduction

1.1. Background

The construction period of the cable structure is short and the shape plasticity is strong. The cable member only bears the axial tension, the material strength utilization rate is high, and the structural force is reasonable. Cable structure is widely used in large-span engineering [1]. The cable structure is mostly used in public buildings such as stadiums and exhibition centers. Taking the cable component as the key stress component, the structural systems such as cable truss [2], cable dome [3] and chord-supported structure [4] are formed. Due to the change of building performance and structural safety performance, the cable structure often needs to be reformed and dismantled. The cable structure has many components and complex forces, which leads to complicated construction process, difficult coordination and high safety risk [5,6]. In the process of structural demolition, due to the unreasonable resource allocation and construction scheme and the low degree of information sharing, it is easy to lead to the occurrence of construction safety accidents, which seriously threatens the life safety of construction personnel

* Corresponding author at: Faculty of Architecture, Civil and Transportation Engineering, Beijing University of Technology, Beijing 100124, China.
E-mail addresses: guoliangshi@emails.bjut.edu.cn (G. Shi), liuzhansheng@bjut.edu.cn (Z. Liu), dechun@bjut.edu.cn (D. Lu), qwzhang2023@bjut.edu.cn (Q. Zhang).

https://doi.org/10.1016/j.istruc.2024.107425
Received 19 May 2024; Received in revised form 21 September 2024; Accepted 27 September 2024
Available online 5 October 2024
2352-0124/© 2024 Published by Elsevier Ltd on behalf of Institution of Structural Engineers.

[7]. Therefore, it is of great significance to realize the efficient scheduling of resources in the demolition process, the optimization of the demolition scheme and the accurate analysis of the safety state.

In the process of structural construction, the cable force is an important basis for measuring the safety state [8]. It is necessary to clarify the development of each cable force to ensure that the cable force can develop smoothly and evenly during the demolition process [9]. In the process of demolition, according to the development of cable force, the best demolition scheme needs to be obtained, and the efficient guidance of the construction process are the focus of this study. In this process, it is necessary to clarify the batching, grading and cable relaxation degree at all levels of the demolition process [10]. Many scholars have proposed new methods and techniques in structural optimization. Saleh et al. [11] proposed an optimization method for the mechanical properties of shear members based on topology optimization. Through numerical analysis, it was found that the optimized connection showed superior performance, and showed higher shear strength and energy dissipation capacity under monotonic and cyclic loading. Ghabussi et al. [12] used steel curved dampers to optimize the seismic performance of symmetrical and single-slope portal frames of variable-slope roofs. Based on the finite element (FE) analysis method, the influence of the curve damper on the energy dissipation, frame strength and stiffness of the portal frame is evaluated. Ghabussi et al. [13,14] used the FE analysis method to evaluate the seismic performance of the cold-formed hot-rolled steel wall system with curved steel dampers. Considering the strain energy, geometric comprehensive index and ultimate bearing capacity, Cao et al. [15] optimized the design of free-form cable--supported reticulated shell. The FE analysis verifies that the designed structure has better mechanical properties and mesh quality. The above optimization methods provide reliable technical support for the optimization of mechanical properties of steel structures and composite structures. Moreover, most of the above optimization methods use the FE numerical simulation method, which also provides a reference for the optimization of the structural construction scheme. In the analysis of construction scheme, the traditional analysis method is to set the boundary conditions and working conditions in the FE model [16], and then obtain the change trend of cable force in the process of structural demolition. There are many uncertain factors such as temperature and wind load in the process of structural demolition. The traditional analysis method cannot accurately obtain the best demolition scheme. In addition, the FE analysis method usually obtains the mechanical response of the structure according to the specific construction instructions to judge the rationality of the scheme. This method is discrete and cannot realize the continuous dynamic analysis of the construction process. Therefore, the optimal selection of construction scheme cannot be realized in the process of finite element calculation. FE analysis focuses on the mechanical information of the structure. However, in the actual construction process, how to realize the efficient allocation of resources in the construction process is the key to guiding the construction. Therefore, the construction process should be further visualized on the basis of FE calculation. The tooling configuration, action position and specific process parameters of each construction step are clarified. The process operation in the process of structural demolition often relies on empirical knowledge and subjective judgment, or needs to manually check the existing construction standards or specifications. Therefore, the current technical methods lead to the low level of construction management, which seriously affects the efficiency and accuracy of demolition construction.

Under the background of the rapid development of the new generation of information technology, intelligent construction is reshaping the mode of MPs analysis and construction scheme control [17]. Through the cross-integration of new generation information technologies such as digital twin (DT), artificial intelligence, building information modeling (BIM), and Internet of Things, solving various problems in engineering construction has become the focus of realizing the intelligence of the construction industry [18–20].

DT is a technology that makes full use of models and their data to achieve real-time synchronization, virtual-real mapping, and digital-analog linkage [21]. As an artificial intelligence method that can learn the deep features of data, deep learning can directly extract high-level features from the original data for perceptual decision-making and efficiently process state data [22]. The combination of sample data and deep learning, that is, the combination of knowledge segmentation method and data-driven deep learning, is conducive to accurate acquisition of optimized construction schemes [23,24].

1.2. Research significance and article structure

The current research results for the analysis of cable structure demolition are relatively scattered. Many scholars pay attention to the collapse resistance of cable structures after members failure [25]. There is a lack of systematic and unified modeling simulation and application method system. The following questions need to be further answered for the transformation and demolition of the cable structures. (1) How to quickly establish a simulation model in the information space to realize the unified expression and virtual reconstruction of the actual demolition process of the cable structure. (2) How to establish a suitable simulation mechanism and simulation strategy to drive the virtual model to accurately simulate and analyze the actual demolition behavior. (3) How to integrate the virtual and real data of virtual space and physical space to realize the closed-loop control of the scheme optimization and structural safety state of the demolition execution process in virtual space. (4) Deep learning algorithms can usually establish regression prediction models, and there is a lack of research on the optimal selection of results.

Aiming at the demolition process of cable structure, this study proposes a DT evolution model-driven optimization method for cable structure demolition. Firstly, the current research status of cable structure demolition, construction scheme optimization and DT is fully investigated, and the support technology for cable structure construction scheme optimization is obtained. Combined with the characteristics of the virtual and real mapping of DT, the evolution process of digital twins in cable structure demolition is clarified. Taking the actual demolition as the core, the visual model, simulation model and data model are established. The mapping mechanism of three types of model information is proposed. The optimal scheme is obtained by systematically analyzing the demolition execution process. Based on the analysis of theory and method, the DT platform is developed to guide the demolition process efficiently and accurately. Establishing a data model to optimize the demolition plan is the core link. The expression method of the relationship between the demolition scheme and the mechanical properties is proposed. The mechanical properties corresponding to various demolition schemes are obtained through the simulation model, which provides a reliable sample for the establishment of the data model. The activation function is introduced to improve the PSO-BP neural network, and the mechanical properties corresponding to the demolition scheme are obtained accurately and quickly. According to the performance law of mechanical properties, the best demolition scheme is obtained. Taking the demolition process of the cable truss experimental model as an example, the digital twin platform is developed to improve the sharing efficiency of construction information and realize the efficient guidance of the demolition process. The main contributions of this study are as follows:

(1) The DT process of cable structure demolition is clarified, and the unified expression and virtual reconstruction of the construction process in the information space are realized. In the process of evolution, a multi-dimensional mapping model of virtual and real is established, which significantly improves the level of construction closed-loop control.
(2) The expression method of the relationship between the demolition scheme and the mechanical properties is proposed. Based on

the improved PSO-BP neural network, the mapping relationship between the scheme and the performance is established. Taking the steady development of cable force as the optimization goal, the best demolition scheme is obtained.

(3) Based on the experimental model, the verification and demonstration application of the theoretical method are carried out to provide reference for the actual construction case. During the experiment, the DT control platform is developed by integrating theoretical methods. Combined with multi-model mapping and scheme optimization, the construction is guided efficiently.

In this study, the organizational structure of each section is as follows:

The second section systematically reviews the literature on cable structure demolition, construction scheme optimization method and DT. In the third section, from the perspective of virtual and real mapping, the DT evolution process of cable structure demolition is clarified. The visualization modeling, simulation modeling and data modeling are integrated to realize the virtualization of the physical demolition process. Fully considering the temporal and spatial evolution law of the demolition process, a DT evolution model of the demolition process is established. Starting from the information flow, component flow, control flow and their relationship, the problems existing in the process of demolition execution are analyzed.

In the fourth section, driven by the twin evolution process, a data mapping method for the demolition process is proposed. The safety risk is characterized by the MPs of the structure, and the expression language of the data association relationship is established. The particle swarm optimization back propagation neural network (PSO-BPNN) is improved, and an intelligent analysis method of MPs in the whole process of demolition is proposed. The best solution is obtained by integrating the changes of MPs between each demolition step. The information of the demolition evolution process is divided into functional modules, and the functional framework of the DT platform is built to efficiently guide the actual construction.

The fifth section is the experimental verification of the theoretical method.

The sixth and seventh sections are the discussion and conclusion of the article.

2. Literature review

This study focuses on the scheme optimization and construction guidance of the demolition process of cable structure. According to the research background and the key problems to be addressed, this section conducts literature research from three aspects: cable structure demolition, construction scheme optimization and DT. On the one hand, the key technologies and shortcomings of the current cable structure demolition are obtained. On the other hand, based on scheme optimization and DT, the method to optimize the cable structure demolition scheme and guide the construction efficiently is found.

2.1. Demolition of cable structure

With the development of urbanization, some completed buildings need to be demolished due to insufficient function and expiration of service life [26]. In the process of removing the cable structure, many scholars have analyzed the influence of the removal of different types of cable members on the MPs of the structure. Yang et al. [27] studied the demolition method of spoke cable-strut tension structure. The effectiveness of the proposed new dismantling method is verified by experiments, and the influence of the dismantling of the struts on the MPs of the cable is obtained. Liu et al. [28] proposed a new analysis method for the influence of component failure on the MPs of spoke cable truss structures. This study accurately obtained the internal force redistribution and reliability of the structure after the failure of the radial cable and the circumferential cable. Zhao et al. [29] studied the MPs of suspend-dome after cable fracture. The study obtained the cable breaking path and the collapse resistance of the remaining structure. Liu et al. [30] established the calculation formula of progressive collapse bearing capacity and deflection of truss string structure under the condition of cable fracture. This study accurately and effectively evaluates the MPs of the structure after component removal, and greatly improves the computational efficiency. Chen et al. [31] analyzed the collapse resistance of double-layer grid structure under different demolition load schemes. Through FE calculation and experimental verification, the internal force redistribution law of the structure is obtained.

For the demolition analysis of cable structures, scholars have focused on the influence of component failure and cable fracture on the MPs of structures. According to the technical regulations of cable structure [32], in the normal use stage and the demolition construction process, the structure should meet the stress requirements and ensure the reliability of the structure. During the service period of the structure, the cable stress does not exceed 0.4 times of the ultimate tensile strength. The maximum displacement of the nodes should not exceed 1/250 of the structural span. In the process of dismantling the structure, the stable development of the cable force should be ensured, and the gap between the cable forces should not be too large. However, there are few studies on how to solve the problem of high safety risk and low efficiency in the demolition process due to the cumbersome structure and difficult coordination. The traditional FE analysis method needs to modify the parameters such as boundary conditions, which leads to a certain lag in the information interaction of the demolition process, which is not conducive to engineering application. In addition, the FE analysis method cannot realize the continuous dynamic analysis of the construction process. Due to the lack of analysis of the mapping relationship between each construction scheme and performance parameters, the effective optimization of the construction scheme cannot be realized. Therefore, it is urgent to establish an effective optimization method for the demolition process scheme, so as to realize the efficient resource scheduling of the whole demolition process and ensure the reasonable force of the structure.

2.2. Construction scheme optimization method

The optimization of the construction scheme can not only ensure the safety performance of the structure, but also save the cost. Many scholars have applied artificial intelligence algorithms to obtain the site layout and process planning of the construction process. Wu et al. [33] proposed a space-time planning model for tower cranes, which significantly saves the cost of construction. El-Abbasy et al. [34] developed an automated system that can optimize different construction management scheduling problems. Chen et al. [35] established a construction scheme optimization model based on genetic algorithm, which effectively reduced the probability of rework. Tomczak et al. [36] used PSO-BPNN to coordinate multiple processes in the construction process, effectively shortening the construction period of the project. Based on PSO-BPNN, Ma et al. [37] optimized the construction quality, which effectively reduced the construction error.

Most of the current research on construction scheme optimization focuses on construction path planning and site layout. There is a strong process correlation in the construction process of cable structure. Due to the interaction between multiple construction steps, few studies have comprehensively considered the correlation and mechanical properties to optimize the construction scheme. In this study, the optimal demolition scheme of the structure is obtained by establishing the mapping relationship between the demolition scheme and the MPs. In addition, deep learning algorithms can usually establish regression prediction models, and there is a lack of research on the optimal selection of results. In this study, the activation function is introduced to improve the PSO-BP neural network to establish the optimization model of the demolition scheme. Based on neural network for data processing, the self-renewal of

engineering knowledge is realized, and the accuracy and efficiency of optimization are significantly improved.

2.3. DT

In the construction industry, the application of DT has greatly improved the visualization and efficiency of construction management [38]. DT can realize the integration of modeling technology and data processing technology [39]. Jiang et al. [40] constructed the overall architecture of the construction management and control system driven by BIM and DT. In the construction process, the stress and operation state of the structure were timely understood. Ritto et al. [41] constructed an intelligent analysis model based on DT for structural damage identification. The combination of this model and machine learning provided a reliable reference for related engineering decisions. Li et al. [42] proposed a DT -driven virtual sensor architecture for the construction safety of trailing suction hopper dredgers. In the virtual model, the operation indexes of mechanical equipment were predicted and evaluated. The virtual sensor assisted the physical sensor to diagnose accurately and timely, and further warned of potential mechanical faults. Teizer et al. [43] outlined the concept of construction safety DT (DTCS) and defined three basic steps in the DTCS workflow. DTCS should be considered as a system-based approach to improve the overall safety performance of the construction process. Wang et al. [44] proposed an interactive and immersive process-level DT system in virtual reality, which integrated visualization and supervision, task planning and execution, and two-way communication to achieve human-machine collaborative construction. Zhang et al. [45] used DT to expand the existing level of detail of the BIM for construction site management. The proposed framework can support construction site monitoring and management, improve quality and efficiency, and improve construction safety.

DT provides model support for dynamic simulation and data extraction of the whole process of demolition. Driven by DT, this study establishes a virtual model of the actual demolition process to realize dynamic simulation and optimal control of the physical space. In addition, the virtual-real mapping and construction scheme optimization are integrated to develop a DT control platform to efficiently guide the whole construction process.

3. DT evolution process of cable structure demolition

In the process of dismantling the cable structure, the MPs are used as the index to judge the feasibility of the scheme [46]. According to the characteristics of virtual and real interaction of DT, the twin evolution modeling methods of physical process virtualization and demolition process are analyzed. Taking the actual demolition as the core, the visual model, simulation model and data model are established. The optimal scheme is obtained by systematically analyzing the demolition execution process. Based on the analysis of theory and method, the DT platform is developed to guide the demolition process efficiently and accurately. The DT optimization and control framework for structural demolition is shown in Fig. 1.

3.1. Virtualization of physical demolition process

Fig. 2 describes the physical space and virtual space of the demolition process. The demolition process of cable structure is a hierarchical transformation system from component unit, equipment unit to the whole structure. As shown in Fig. 2(a), the structural member units and equipment units are classified as field control layers. The specific demolition process acts on a specific component unit by assigning construction information to the equipment unit. In the process of demolition, the MPs of the structure are the key indicators to judge the rationality of construction [47]. In the twin system, on the one hand, the measured data of the site are collected, and on the other hand, the simulation data are compared to ensure the construction accuracy [48]. For the whole structural system, the stress state of the components in each demolition stage is optimized. Therefore, in the actual demolition process, unit level, system level and structure level are divided from micro to macro. In order to facilitate the expression of virtual space, the field control layer, data acquisition and monitoring layer and optimization layer are defined from three levels. As shown in Fig. 2(b), three axes are divided based on the three requirements of DT: presentation, mapping and optimization. The three axes describe the physical space from three dimensions: visual model, simulation model and data model. From the axial point of view, three different dimensions realize the mapping of all levels of physical space. Concomitantly, different scales are divided within each dimension to form three closed loops. For example, starting from a single axis, the visualization of objects, indicators and processes characterizes the control layer of the site and acts on the specific demolition component unit. From the perspective of the ring, taking the basic component unit as an example, the component modeling is carried out in the visual model. The MPs of the basic units in the simulation model are analyzed by assigning physical properties to the components. The component parameters are extracted and the stress

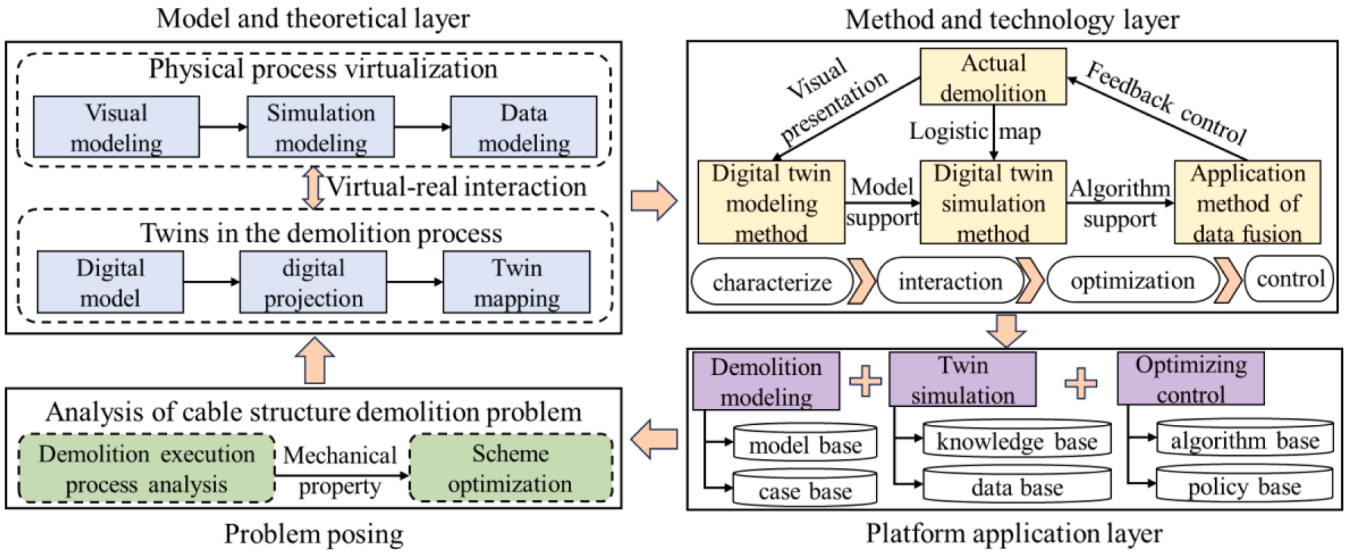


Fig. 1. DT optimization and control framework for structural demolition.

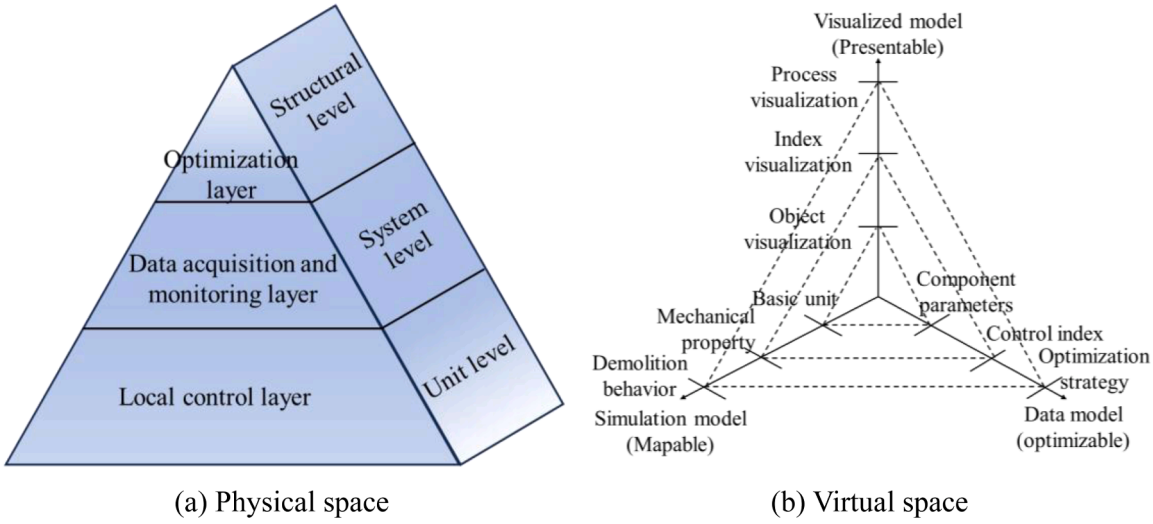


Fig. 2. Virtualization of physical demolition process.

state of the structure is evaluated in the data model [49]. Finally, the MPs of the components are characterized in a visual form.

In the process of dismantling the cable structure, it is necessary to clarify the specific process parameters of the construction process, such as construction location, mechanical control index and construction equipment. Virtual-real mapping is the guarantee of construction optimization. Therefore, the visualization model, simulation model and data model of mapping physical entities are established in virtual space. The visualization model depicts the geometric properties of the structure and clarifies the components and equipment involved in the construction. The simulation model is based on the FE analysis method to obtain the structural mechanics control indexes corresponding to each construction scheme, which provides data support for the scheme optimization. In order to obtain the best construction scheme quickly and accurately, a data model is established to form the mapping relationship between the construction scheme and the mechanical index. By analyzing the change of mechanical properties, the best construction scheme is obtained, and finally the specific process parameters of the best scheme are output through the visual model to guide the whole construction process efficiently.

(1) Visual modeling.

Through the visualization model, a virtual space that is highly consistent with the physical space in visual effects is constructed. The visual model provides a link of virtual and real interaction for the optimal control of the demolition process, reflecting the 'presentability' of DT. In the visualization model, it is divided into three progressive levels: object visualization, index visualization and process visualization. Among them, the visualization of objects and indicators describes the static scene of virtual space, and the visualization of process constructs the dynamic operation process of virtual space.

1) Object visualization. In order to realize the accurate characterization of structural components and demolition equipment, the visualization of objects from two aspects of geometric attributes and operational attributes is proposed. Geometric attributes include the type and spatial location of components and devices. The operation attributes include the logical relationship between the steps in the demolition process and the operation process of the equipment.
2) Indicator visualization. The MPs of the structure are the key indicators of demolition control. By analyzing the stress state of each component during the demolition process, the process attributes are expressed in a visual form.
3) Process visualization. By using three-dimensional animation technology and virtual roaming technology, the scene reproduction of site layout, component demolition process and equipment transfer process is realized.

(2) Simulation modeling.

The simulation model includes the modeling and analysis of the basic units, MPs and demolition behavior of the physical space. By simulating the MPs of the structure under various random disturbances and construction schemes, the prediction of the demolition behavior is realized to assist the optimization decision, which reflects the ' mapability' of the DT. The modeling of the basic unit depicts the static structure. The analysis of MPs and demolition behavior depicts the relationship between the elements of the dynamic demolition process.

1) Basic unit modeling. In the simulation model, the FE model of the structural member is established with the action object of the demolition process as the core. Based on this, the MPs of the structure under various stages, various schemes and various disturbances are analyzed.
2) MPs analysis. On the basis of structural modeling, the stress state of various components is obtained. Taking the MPs of the components as the evaluation index, the feasibility of the demolition scheme is judged. The analysis of MPs lays a model foundation for subsequent demolition task scheduling and equipment turnover planning.
3) Demolition behavior analysis. Based on the basic unit modeling and MPs analysis, the demolition behavior is analyzed by dynamically associating the construction process. The simulation analysis is carried out for the optimal demolition scheme. In the visualization model, the process visualization realizes the orderly turnover of the equipment in the virtual space, and depicts the demolition operation mechanism.

(3) Data modeling.

Data model is the core of driving demolition optimization. The control of virtual space to physical space is realized by constructing parameter model, control index model and optimization strategy model, which reflects the 'optimization' of DT.

1) Component parameter model. In data modeling, the expression of parameter correlation is the basis of scheme optimization. Taking the MPs as the driving force, the relationship between the basic parameters (position, load, prestress, etc.) and the mechanical parameters (cable force, displacement, etc.) is mainly investigated.
2) Control index model. The MPs of the structure are obtained in the simulation model. Optimized samples are established by parameter

correlation. Driven by deep learning, an analysis model of control indicators is established to accurately and quickly obtain the structural MPs corresponding to various demolition schemes.

3) Optimization strategy model. According to the set target, the demolition scheme is adjusted, and the decision-making basis is the MPs of the structure, which ultimately acts on the basic unit. During the demolition process, the target set can be expressed as: $Set_T = \{\text{best performance}, \text{lowest cost}, \text{highest efficiency}\}$. The optimization strategy is represented in a visual form to guide the actual demolition process.

3.2. DT evolution of demolition process

Based on the virtual modeling of the physical demolition process, the twin model simulation, analysis, optimization and control are carried out. The DT evolution model is the engine that drives this process (as shown in Fig. 3). The DT evolution model includes four aspects: basic model, correlation relationship, fusion process and evolution process. The basic model includes physical entity (actual demolition) and virtual model (visual model, simulation model and data model). Physical entity is the object of modeling, analysis, optimization and control. The three sub-models in the virtual model realize the mapping of physical entities from different dimensions such as visual presentation, simulation, and optimization decision. The evolution process divides the application process of DT into three progressive stages: digital model, digital projection and twin mapping. In the spatial dimension, the relationship represents the mapping relationship between the virtual model and the physical entity. The fusion process describes the typical applications between virtual models and physical entities in different evolution stages from the application dimension.

By integrating physical entities, visual models and simulation models, a digital model space can be constructed and applied to the simulation analysis before demolition and the verification of construction schemes. Based on the digital model, the integrated data model can construct the digital projection space, and realize the automatic collection of physical space construction information and the monitoring and control in the process. On the basis of digital model space and digital projection space, the demolition scheme is optimized based on historical and real-time data, and the feasibility and scientificity of demolition are continuously improved. Through the integrated simulation and visualization model, the optimization results are directly fed back to the actual demolition process.

The DT evolution model of the demolition process ensures the accuracy and efficiency of the actual construction process. From the time dimension, the model describes the closed-loop process from the understanding of the demolition process (digital model), to the perception of the construction process (digital projection), to the optimization of demolition behavior based on data simulation and mining (twin mapping). From the perspective of spatial dimension, the model characterizes the fusion process of physical space and virtual space from relative separation, one-way interaction to two-way interaction. From the perspective of application dimension, the model depicts the whole application process of actual demolition from simulation planning to real-time monitoring and then to optimal control.

3.3. Analysis of the execution process of cable structure demolition

There are three closed loops of component flow, information flow and control flow in the demolition process of cable structure. The efficient development of the demolition process depends on the precise control of the MPs of the components and the unimpeded construction information. Therefore, starting from the component flow, information flow, control flow and their relationships, the execution process of cable structure demolition is systematically analyzed, as shown in Fig. 4.

According to the temporal and spatial evolution law of the demolition process, the information is transmitted to the component flow through the DT platform in the form of process flow, force cloud diagram and operation procedure to control the demolition execution process.

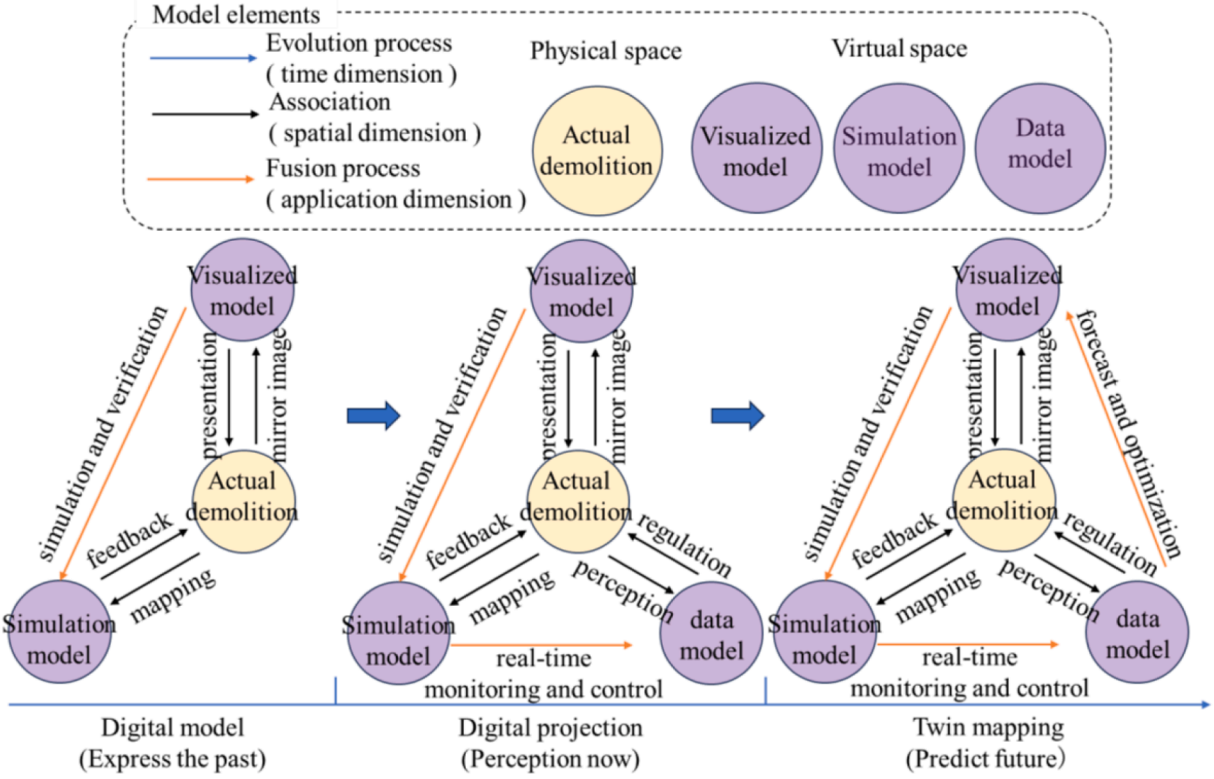


Fig. 3. Composition and application of each evolution stage of DT in demolition process.

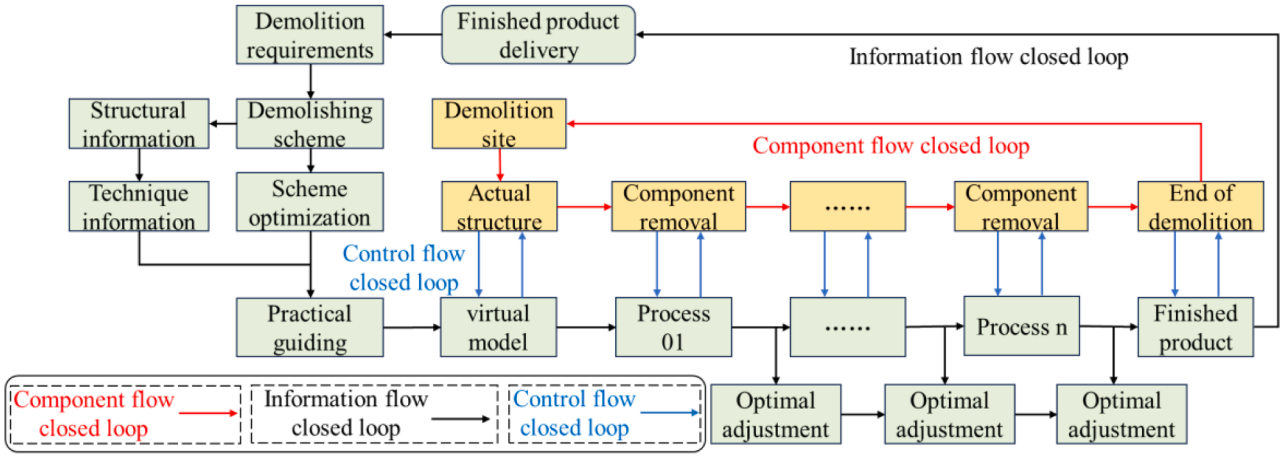


Fig. 4. Demolition process information flow, component flow, control flow and their relationship.

The components are removed in an orderly manner through construction equipment under the action of information flow, and the MPs are fed back to the information flow. According to the mechanical information in the information flow, the demolition scheme is optimized and the construction process is guided to form a control flow closed loop.

Before the implementation of the demolition, the information flow is dominant. The visual model and the simulation model are established in the virtual space to obtain the mapping relationship between the demolition scheme and the MPs in advance. The training samples in the data model are used to optimize the demolition scheme. By analyzing the MPs corresponding to various demolition schemes, an effective, accurate and reliable control flow is formed to optimize the decision-making. In the demolition execution, the component flow is dominant. The optimized demolition scheme is applied to the structural members. The measured values and theoretical values of the MPs of each process in the demolition process are compared in real time to ensure the consistency of the construction.

4. Data mapping of demolition process

By analyzing the DT evolution process of cable structure demolition, MPs are the key indicators to judge the feasibility of the scheme. Establishing a data model to optimize the demolition plan is the core link. In this section, the expression method of the relationship between the demolition scheme and the MPs is first proposed. The MPs corresponding to various demolition schemes are obtained through the simulation model, which provides a reliable sample for the establishment of the data model. The PSO-BPNN is improved to accurately and quickly obtain the MPs corresponding to the demolition scheme. According to the performance law of MPs, the best demolition scheme is obtained. Finally, a DT platform is established to guide the actual demolition process.

4.1. Expression of data association

The demolition process of the cable structure is divided into two parts, namely, roof load demolition and cable relaxation. Establishing the expression method of the relationship between the demolition scheme and the MPs is the basis for optimization. The sudden change of cable force between each step and the large gap of cable force between cables seriously affect the safety of the structure. In the process of removing the roof load, the change rate of the cable force in each demolition step is taken as the optimization goal. For the *ith* demolition step (PU_i), it is necessary to establish the relationship between the load condition (CB_i) and the cable force (F_i), which is expressed as Eq. (1). The optimal demolition scheme (CPU) is obtained by minimizing the cable force change of adjacent demolition steps, which is expressed as Eq. (2).

$$PU_i = (CB_i|F_i) = \left(\begin{array}{cccc|c} a_1^{m_1} & a_1^{m_2} & \cdots & a_1^{m_p} & F_1 \\ a_2^{m_1} & a_2^{m_2} & \cdots & a_2^{m_p} & F_2 \\ & & \cdots & & \cdots \\ a_n^{m_1} & a_n^{m_2} & \cdots & a_n^{m_p} & F_n \end{array}\right) \tag{1}$$

$$CPU = \bigcup_{i=1}^{i=q}(PU_i - PU_{i-1}) \Rightarrow^{\text{target}} \min(\Delta F_i) \tag{2}$$

In the Equations, $a_i^{m_j}$ represents the load of the *jth* node on the *ith* cable member, and the specific values are represented by 0 and 1. After the node load is removed, the value is 0, and the node load is not removed, the value is 1. In the optimization process, the cable force change (ΔF_i) of the adjacent demolition steps ($\bigcup_{i=1}^{i=q}(PU_i - PU_{i-1})$) is comprehensively considered, and the minimum change rate of each cable force is taken as the optimization objective.

In the process of cable relaxation, the change rate of cable force in each relaxation step and the difference of cable force between cable components are taken as the optimization objectives. For the *ith* relaxation step (PR_i), it is necessary to establish the relationship between the relaxation degree (T_j) and the cable force (F_j), which is expressed as Eq. (3). By minimizing the cable force change of the adjacent relaxation steps and the cable force gap between the cables, the optimal demolition scheme (CPR) is obtained, which is expressed as Eq. (4). In the process of relaxing the cable, the prestress is adjusted by applying the element temperature load, which is expressed as Eq. (5).

$$PR_i = (T_j|F_j) = (\ t_1 \quad t_2 \quad \cdots \quad t_n \ |\ F_1 \quad F_2 \quad \cdots \quad F_n\ ) \tag{3}$$

$$CPR = \bigcup_{i=1}^{i=p}(PR_i - PR_{i-1}) \overset{\text{target}}{\Rightarrow} \begin{cases} \min\ (\Delta F) \\ \min\ (\Delta F^*) \end{cases} \tag{4}$$

$$\Delta L = \alpha t L \tag{5}$$

In the Equations, t_i denotes the temperature load applied by the *jth* cable in the *ith* relaxation step. The value is calculated by Eq. (5). α represents the linear expansion coefficient of the cable. L equals to the length of the radial cable. ΔL indicates the length variation of the cable in each relaxation process. In the optimization process, the cable force variation of the adjacent relaxation step ($\bigcup_{i=1}^{i=p}(PR_i - PR_{i-1})$) is comprehensively considered, and the minimum cable force change rate (ΔF) and the minimum cable force difference between each cable (ΔF^*) are taken as the optimization objectives.

4.2. MPs analysis and scheme optimization of demolition process

Various demolition schemes and their corresponding MPs are obtained in the simulation model. Data samples are formed according to the expression rules of the association relationship in Section 4.1. Driven by the improved PSO-BPNN, a data model is established to obtain the optimal demolition scheme. The process of analysis and optimization is shown in Fig. 5.

The traditional neural network obtains the output factor according to the input factor, and ensures a high degree of fit between the predicted value and the true value. The PSO-BPNN can output the predicted value accurately and efficiently after sample training. However, in the optimization process of the demolition scheme, not only the specific value of the cable force in each step is required, but also the cable force change rate between the various demolition steps and the cable force gap between the cables need to be obtained. The demolition scheme is selected under the goal of minimum cable force change rate and minimum gap. In this study, the activation function is introduced to improve the PSO-BPNN, seen in Eq. (6).

$$\Gamma(x) = \alpha \tanh(x) = \alpha \frac{e^{x} - e^{-x}}{e^{x} + e^{-x}} \tag{6}$$

In the Equation, x represents the change rate of cable force and the gap value of cable force obtained by neural network analysis. The value range of $\tanh(x)$ is $[-1,1]$. According to the optimization conditions, the value range of $\tanh(x)$ is enlarged and narrowed. By introducing an activation function, the samples on the target interval are quickly selected. Finally, the best demolition scheme is selected according to the MPs of the sample.

According to Reference [50,51], the main factors affecting the accuracy of neural network analysis are the number of hidden layer neurons and the number of iterations. The number of neurons in the hidden layer is determined by Eq. (7) [52,53]. In the neural network, the load and relaxation of the cable (represented by temperature) are taken as the input layer, and the cable force change of the cable is taken as the output layer.

$$l = \sqrt{n+m} + a \tag{7}$$

Among them, l is the number of neurons in the hidden layer. n equals to the number of neurons in the input layer, that is, the bearing capacity of each cable. m denotes the number of neurons in the output layer, that is, the response of the cable force. a is a constant between [1,10]. The fitting degree between the predicted value of the neural network and the simulated value of the FE analysis is used as the index to evaluate the neural network. The calculation method is expressed as Eq. (8).

$$R^2 = 1 - \frac{\sum_{1}^{N} \left(y_{MVi} - y_{TVi}\right)^2}{\sum_{1}^{N} \left(y_{MVi} - \overline{y_{MV}}\right)^2} \tag{8}$$

In the Equation, R^2 represents the degree of fitting. N equals to the number of samples. y_{MVi} is the simulation model simulation value of MPs, and $\overline{y_{MV}}$ denotes the average value of the simulation model simulation value of MPs. y_{TVi} represents the neural network prediction value of MPs.

4.3. Functional architecture and construction guidance of DT platform

Driven by the DT evolution model, the visualization model, simulation model and data model of the demolition process are established in

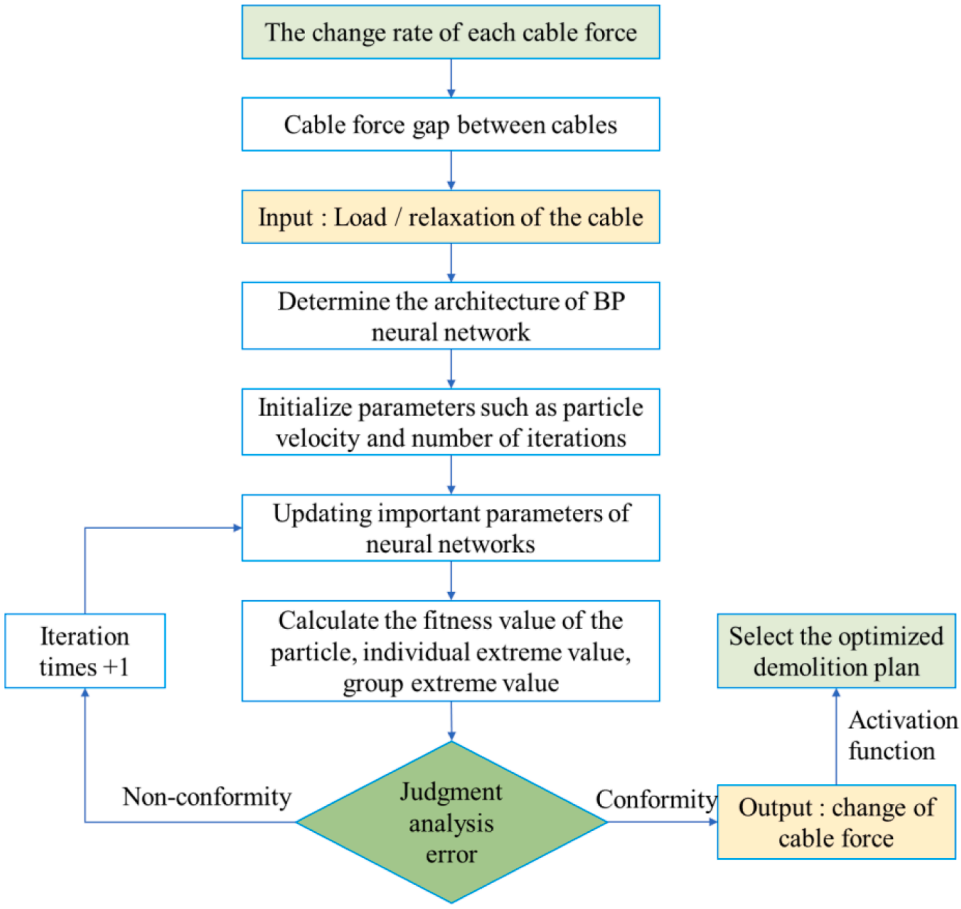


Fig. 5. Demolition scheme analysis and optimization process.

the virtual space. Taking the MPs of the structure as the evaluation index, the best demolition scheme is obtained. In order to further accurately guide the construction, the functional architecture of the DT platform is constructed. The twin platform integrates three functions of demolition modeling, twin simulation and optimal control to form the corresponding model library, instance library, knowledge library, database, algorithm library and strategy library. Based on the visual model, the demolition modeling function is realized to monitor the real-time stress state of the actual structure. Based on the simulation model, the twin simulation function is realized, the feasibility of the demolition scheme is analyzed and the corresponding MPs are obtained, which provides reliable data samples for the optimization of the scheme. Driven by the data model, the optimal control function is realized, the best demolition scheme is obtained and the operation instructions are issued for the actual demolition. By comparing the measured values and theoretical values of the MPs of the demolition process in the platform, the actual operation scheme is corrected to ensure the consistency of the construction. The establishment and application of twin platform effectively avoid the construction risk. The operation process of the DT platform in the demolition process is shown in Fig. 6.

5. Case study

In order to verify the feasibility of the proposed theoretical method, a DT platform is developed to guide the demolition process based on the cable truss experimental model. The visualization model, simulation model and data model are integrated in the twin platform. The optimized demolition scheme is output in the platform, which accurately and effectively guides the whole construction process.

5.1. Experimental design

The cable truss span of the experimental model is 6 m, which is composed of 12 cable trusses and inner ring beams, flying columns, outer ring beams and outer columns. The experimental model and visual model are shown in Fig. 7. The cable adopts the sealing cable with tensile strength of 1670 MPa, The elastic modulus is 2.05×10^{11}Pa. And other components adopt Q345B steel. In the analysis of the construction process of the structure, the cable member adopts the linear elastic material model, and the steel member adopts the elastic-plastic material model [54]. According to previous study [55], in the design process of the experimental model, the influence of cable size, initial tension and other factors on the internal force and configuration of the structure is mainly investigated. Through parameter analysis and optimization selection, the best design parameters are obtained to ensure the rationality of the structural force. Moreover, in Reference [56], the mechanical response analysis of various static and dynamic loads is carried out, which meets the requirements of structural performance. The radial cable is divided into two layers, namely the upper and lower radial cable (URC, LRC). The diameter of the radial cable is 9 mm, and the morphological cable forces of the URC and LRC are 15 kN and 16 kN, respectively. The URC and LRC are connected by two struts. The inner ring beam includes the upper ring beam and the lower ring beam, which are connected by the flying column. The height of the flying column is 0.75 m, and the radius of the inner ring beam is 0.5 m. The height of the outer column is 1.5 m.

According to the characteristics of the experimental model, the demolition process is divided into two steps: the removal of the counterweight block and the relaxation of the cable. In the process of removing the counterweight block, the cable force of each radial cable is obtained according to the load of each node of each cable truss. In the process of removing the counterweight block, the change rate of cable force will affect the deformation of the structure, and there is the safety risk. If the cable force gap is too large, it will lead to the dislocation of the components and lead to serious safety accidents [57]. The best relaxation scheme is obtained by taking the change rate of cable force in each relaxation stage and the cable force gap between each cable as the evaluation index. When the weight block is removed and the cable relaxation is completed, the demolition process of the structure is completed. The arrangement of outer ring beams and columns in the structure is completely symmetrical, and only bears the action of gravity. Therefore, the demolition scheme of the two types of components was not optimized in the experiment.

This study investigates the scheme optimization of the structural

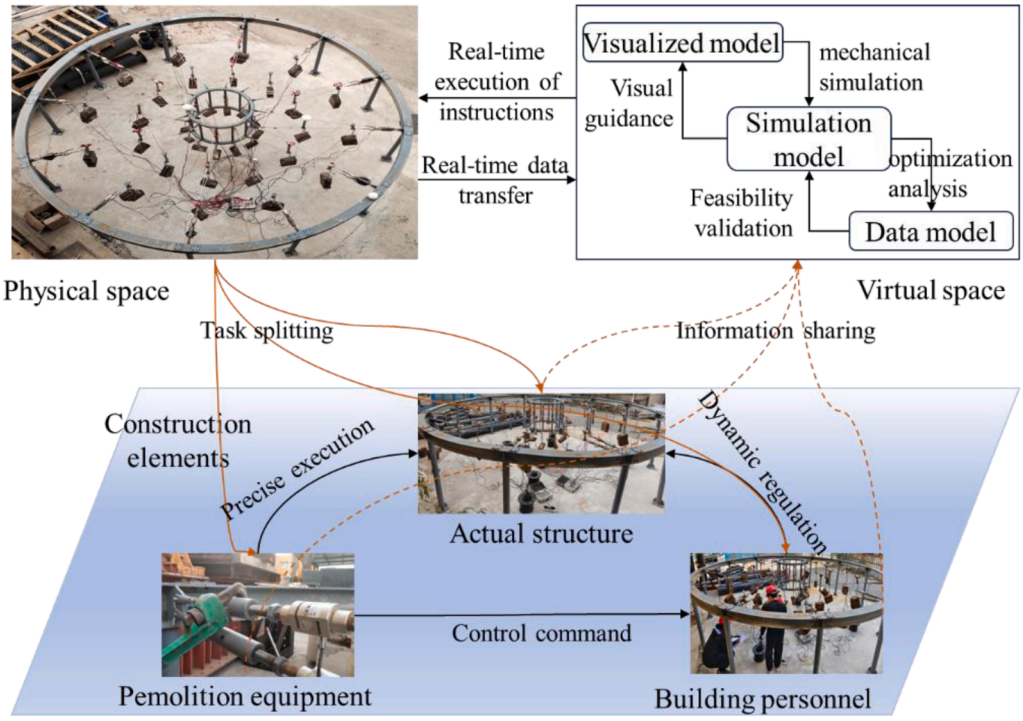
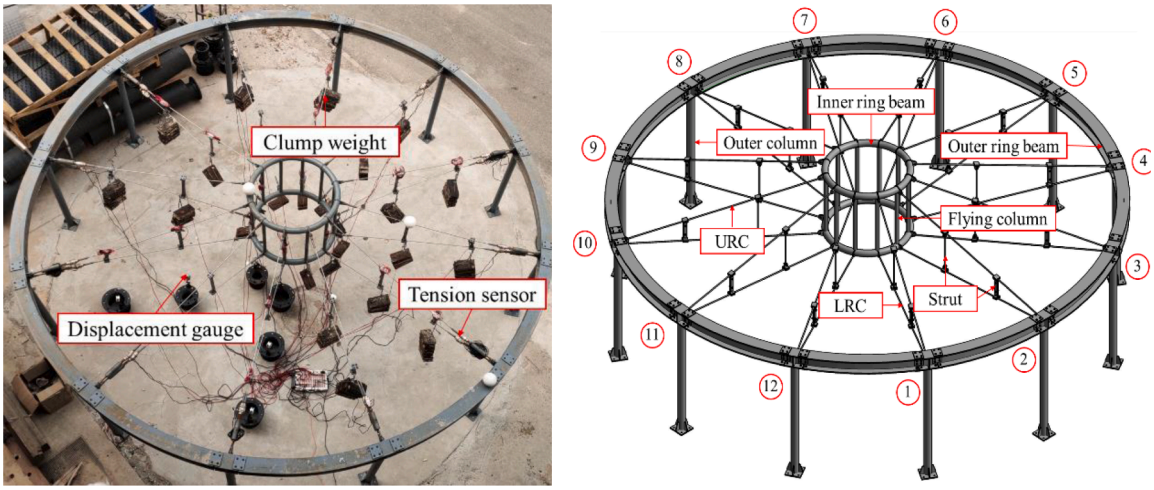


Fig. 6. Operation process of DT platform in demolition process.



(a) experimental model (b) visualized model

Fig. 7. Experimental structure.

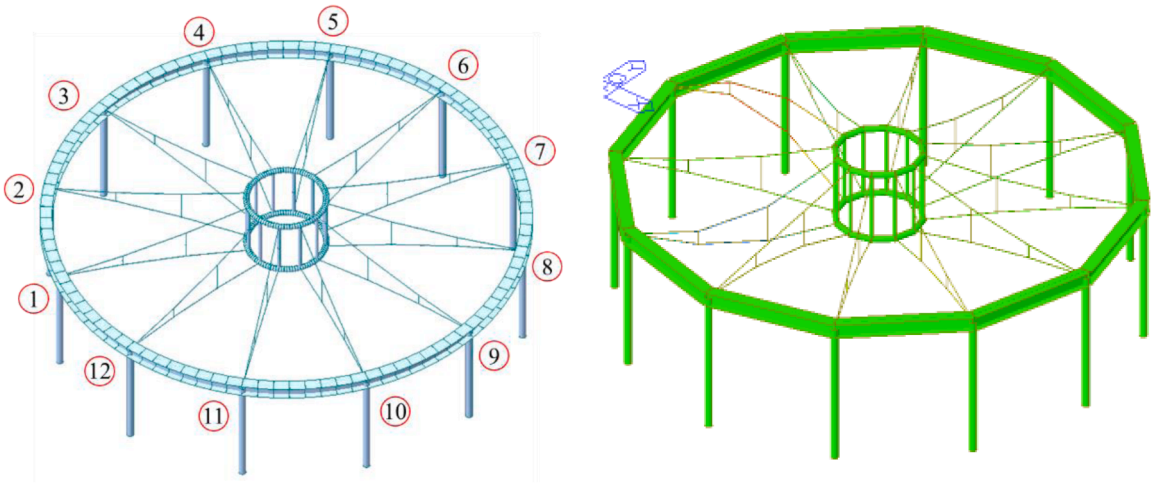
demolition process. In order to conform to the engineering practice, 1.0 times of constant load is applied to the structure. According to the load specification [58], the constant load is 0.15kN/m^2. The surface load is converted into equivalent node load. The node load is replaced by the counterweight block. The node load on the flying column is 33 N, the node load on the inner strut is 92.3 N, and the node load on the outer strut is 146 N.

5.2. Analysis of the MPs of the structure during the demolition process

Based on the visual model, the detailed structure of the structure is presented intuitively. In the visualization model, the positional relationship between the components is clarified. According to the analysis of Section 3, in the process of dismantling the cable structure, the MPs are used as the index to judge the feasibility of the scheme. Based on the FE modeling technology, the simulation model of the structure is established, and the MPs simulation analysis of the demolition scheme is carried out in the model, as shown in Fig. 8.

In the simulation model, the operation command of the demolition process is set up to obtain the change of each cable force. The demolition scheme and the corresponding MPs are organized into an association relationship according to Eqs. (1) - (4) to provide samples for the establishment of the data model. Taking the removal process of the counterweight block as an example, the form of the data sample is shown in Table 1.

In the process of cable relaxation, the change of cable length is adjusted by setting temperature, and the prestress of cable is released. Starting from the design elevation of the ring beam under the structural form, the temperature effect of the radial cable is applied to the outer element. By increasing the length of the cable, the cable force is gradually reduced. The construction process of the cable structure has the characteristics of large deformation and large displacement. Therefore, geometric nonlinearity and the number of iterations should be considered in the simulation process to ensure the accuracy and convergence of the calculation. According to Eq. (5), the relaxation process is simulated by the relationship between temperature and cable length. α is the linear expansion coefficient of the cable. In this study, $\alpha = 1.59 \times 10^{-5}$, l represents the length of the outermost radial cable. In the twin model, the length of the upper radial cable and the lower radial cable is 836.7 mm. The cable force changes corresponding to each construction step are obtained in the twin model. Through a large number of simulation calculations, a sample set between the construction scheme and



(a) simulation model (b) MPs analysis

Fig. 8. Simulation model and MPs analysis.

Table 1
The form of data samples.

Dismantling steps	NO.1 cable truss	NO.2 cable truss	…	NO.12 cable truss	URC1 cable force	URC2 cable force	URC3 cable force
0	1.11	1.11	…	1.11	13.688	13.688	13.688
1	1.10	1.10	…	1.10	14.708	14.708	14.708
2	1.00	1.00	…	1.00	13.502	13.502	13.502
3	0.00	0.00	…	0.00	13.742	13.742	13.742

the mechanical properties is formed, which provides data support for the optimization calculation of the neural network.

In the process of training the neural network, the load of each cable truss node is taken as the input element, and the cable force change of each cable is taken as the output element. The load of each cable truss node is represented by 0,1. For example, 1.10 indicates that there are loads on the fly column joints and the inner strut joints, and the outer strut joints remove the load. Driven by neural networks, a data model is formed. The number of hidden layer neurons and the number of iterations are the key factors affecting the accuracy of neural network analysis [59]. According to Eq. (8), the fitting degree of data analysis is calculated, the effectiveness of neural network is evaluated, and the best parameters are obtained. Through the sensitivity analysis of the two types of parameters, the output values of the neural network and the simulation model are compared, and the fitting degree corresponding to different parameters is shown in Fig. 9. It is found that when the number of hidden layer neurons is 13 and the number of iterations is 400, the fitting degree is 97.5 %. When the number of nodes and the number of iterations are increased again, the increase of the fitting effect is not obvious. After the training is completed, the best parameters of the neural network are shown in Table 2.

Based on the optimal parameters, a highly robust neural network is established to form a data model for the optimization of demolition schemes. By inputting different demolition schemes, the corresponding changes of each cable force can be analyzed with high precision. Under different demolition schemes, the comparison between the predicted value and the analyzed value of each cable force is shown in Fig. 10 and Fig. 11.

In Figs. 10 and 11, the value with * is the predicted value of the data model, and the value without * is the real value in the simulation model. Through comparison, it is found that the output of the data model has high accuracy. The analysis accuracy of various samples is shown in Fig. 12. In the data model, the development of cable force can be obtained according to the load condition, which greatly reduces the workload of simulation analysis. Morever, according to the analysis results of the data model, the sample where the objective function is located can be activated to obtain the best demolition plan.

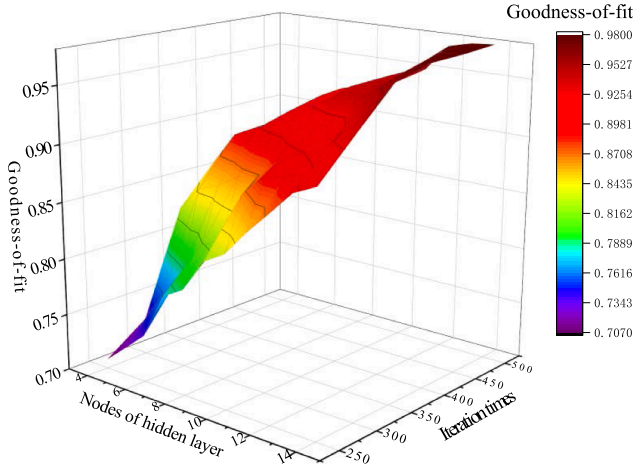


Fig. 9. The fitting degree corresponding to different parameters.

Table 2
Optimal parameters of neural network.

Parameter	Regulating range	Optimal value
Number of hidden side nodes	[0, ∞]	13
Maximum number of iterations	[0, ∞]	400
Learning rate	[0,1]	0.01
Number of particles	[0, ∞]	200

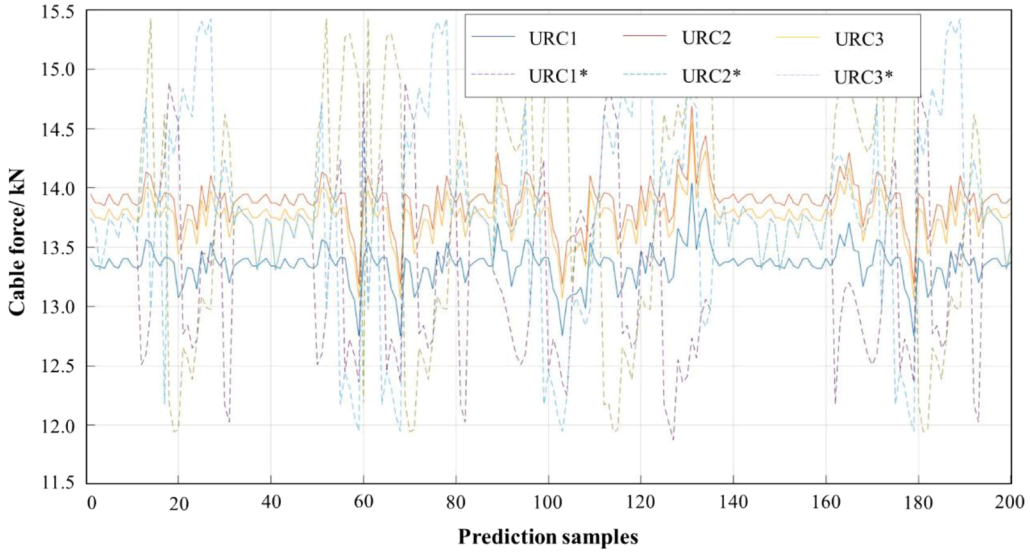
5.3. Best demolishing scheme acquisition

According to the data model, the mapping relationship between the demolition scheme and the change of cable force is formed. The PSO-BPNN is improved by introducing the activation function through Eq. (6). The sample satisfying the objective function in the mapping relationship is selected by the activation function to form the best demolition scheme. In the process of removing the counterweight block, the change rate of cable force between each step is small. Select $\alpha = 0.1$ in the activation function. The optimal demolition scheme of the counterweight block is to dismantle every three pieces, and each piece is dismantled according to the order of the inner brace node, the outer brace node and the flying column node. In the process of removing the counterweight block, a total of 12 steps are divided. In the process of removing the counterweight block, the development of the cable force is shown in Fig. 13.

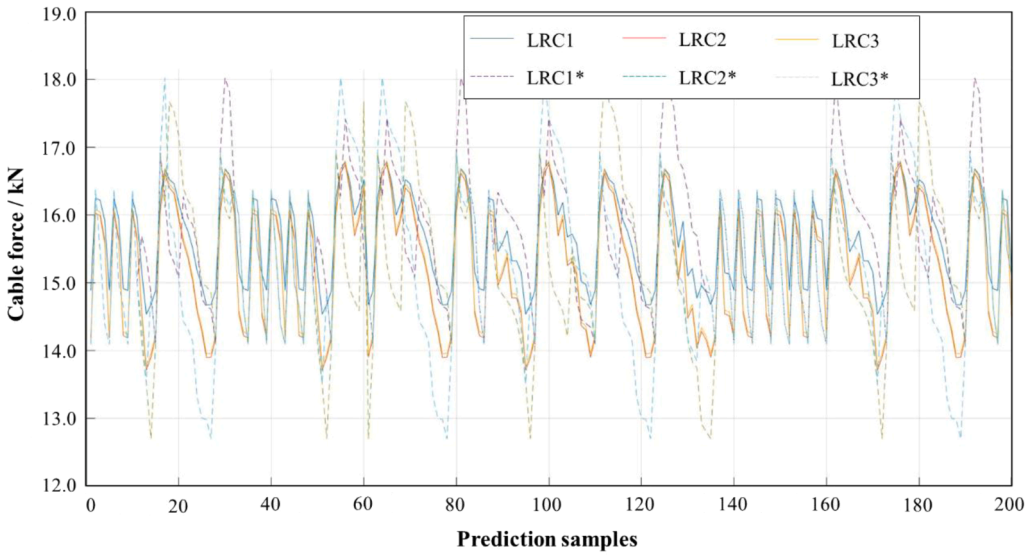
The change rate of cable force is more significant in the process of cable relaxation. Taking $\alpha = 0.5$, the best relaxation scheme is obtained in the data mapping relationship. According to the selected scheme, under the premise of ensuring that the change rate of cable force meets the requirements, the most reasonable force, the most convenient demolition and the most cost-saving scheme are obtained respectively. In the case of considering the reasonable force, the change rate of cable force in each stage is the most uniform, and the corresponding relaxation scheme is the most reasonable. The relaxation scheme is divided into three batches, considering the symmetry, taking the first, fourth, seventh and tenth as the same batch, other batches and so on. The relaxation process is divided into five stages. The first stage relaxes to 80 % of the prestress, the second stage relaxes to 60 %, the third stage relaxes to 40 %, the fourth stage relaxes to 20 %, and the fifth stage relaxes to 0. Thus, 15 relaxation steps are formed. In the process of cable relaxation, the development of cable force is shown in Fig. 14.

In the case of considering convenience, the less the accumulation of the divided relaxation batches and series, the corresponding relaxation steps are the least. The relaxation scheme is divided into four batches. Considering symmetry, the first batch, the fifth batch and the ninth batch are the same batch, and other batches are similar. The relaxation process is divided into three levels. The first level is relaxed to 60 % of the prestress, the second level is relaxed to 30 %, and the third level is relaxed to 0. Thus, 12 relaxation steps are formed, and the cable force corresponding to each tension step is shown in Fig. 15.

In the case of considering the cost, the more the number of slack batches divided means the less the equipment used, the corresponding economy is the strongest. This is the last plan divided into six batches, considering the symmetry, the first and seventh pieces are the same batch, other batches and so on. The relaxation process is divided into three levels. The first level is relaxed to 50 % of the prestress, the second level is relaxed to 20 %, and the third level is relaxed to 0. Thus, 18



(a) Comparison of predicted and analytical values of URC 1-3 cable force



(b) Comparison of predicted and analytical values of LRC 1-3 cable force

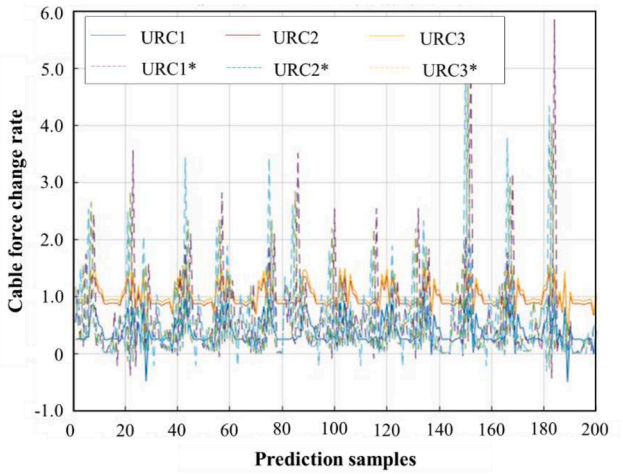
Fig. 10. Comparison between the predicted value and the analytical value of cable force in the process of removing counterweight block (a) Comparison of predicted and analytical values of URC 1–3 cable force (a) Comparison of predicted and analytical values of LRC 1–3 cable force.

relaxation steps are formed, and the cable force corresponding to each relaxation step is shown in Fig. 16.

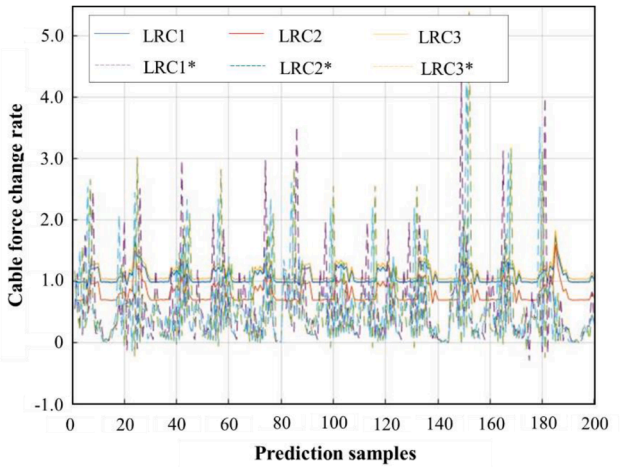
5.4. Demolition process guidance

Based on the cable structure demolition optimization method driven by the DT evolution model, a DT demolition control platform is developed, which provides a tool for the efficient management of the construction process. As shown in Fig. 17, the platform includes three modules: demolition modeling, twin simulation and optimal control to ensure the efficient execution of the demolition process. The visual model of the structure included in the demolition modeling module is previewed in the form of animation, and the feasibility of the optimized construction scheme is analyzed. In the twin simulation module, the visual model is given physical material properties to form a simulation model of the structure. In the simulation model, the demolition operation instruction is set to obtain the MPs data. The optimization control module receives the data samples generated by the twin simulation module, and establishes the data model for the optimization analysis of the demolition scheme driven by the neural network. The optimized demolition scheme is formed and output in this module to accurately guide the whole process of demolition. Based on the optimized demolition scheme, the twin platform outputs the construction sequence of the counterweight block and the cable. The turnover of the demolition equipment is guided in a visual form, which acts accurately on the structural members. Driven by the twin platform, the construction safety and accuracy are ensured, and the construction efficiency is improved.

In the process of demolition, the best scheme is selected for the demolition of the counterweight block. The most reasonable scheme of cable relaxation is selected. According to the optimized demolition plan in Section 5.3, the execution instructions are output in the platform to



(a) The comparison between the predicted value and the analytical value of the change rate of the URC 1-3 cable force



(b) The comparison between the predicted value and the analysis value of the LRC 1-3 cable force change rate

Fig. 11. Comparison of the predicted value and the analytical value of the cable force change rate during the relaxation process of the cable(a) The comparison between the predicted value and the analytical value of the change rate of the URC 1–3 cable force (b) The comparison between the predicted value and the analysis value of the LRC 1–3 cable force change rate.

clarify the turnover and position of various equipment, and accurately guide the whole process of demolition. As shown in Fig. 18, the measured values and theoretical values of the MPs of the demolition process are compared (the construction accuracy is above 95 %). Relying on the DT control platform, on the one hand, the optimal demolition scheme is effectively output to ensure the safety of the structure. On the other hand, the construction resources are efficiently scheduled in the form of visualization, which ensures the construction efficiency. Facing the whole process of demolition, the DT platform effectively realizes the coordination of component flow, information flow and control flow. According to the optimized construction scheme, the cable force of the structure can change smoothly, which avoids the problem of excessive deformation of the supporting members caused by the sudden change of the cable force and the large gap. The development and application of DT platform effectively ensure the safety and construction accuracy of the structure during the construction process.

6. Discussion

The cable structure has the phenomenon of complex component connection, cumbersome demolition process and difficult process coordination, which seriously affects the construction safety and efficiency [60]. The following questions need to be further answered for the transformation and demolition of cable structures. (1) How to quickly establish a simulation model to realize the unified expression and virtual reconstruction of the actual demolition process of the cable structure in the information space. (2) How to establish a suitable simulation mechanism and simulation strategy to drive the virtual model to accurately simulate and analyze the actual demolition behavior. (3) How to integrate the virtual and real data of virtual space and physical space to realize the closed-loop control of the scheme optimization and structural safety state of the demolition execution process in virtual space.

In view of the above problems, this study proposes a cable structure demolition optimization method driven by the DT evolution model. Firstly, the evolution process of DT in demolition is clarified, including the virtualization of physical demolition process, the evolution of DT in demolition process and the analysis of execution process. Taking full account of the temporal and spatial evolution of the demolition process, a DT evolution model is established to further clarify the structural changes and control indicators of the whole construction process. On this basis, the data mapping method of demolition process is proposed to realize the optimization of construction scheme. The safety risk is characterized by the MPs of the structure, and the expression language of the data association relationship is established. The best solution is obtained by improving the PSO-BPNN. The theoretical method is applied to the verification in the experimental model. The experimental results show that this research method can accurately and quickly obtain the best construction scheme, ensure the stable development of structural MPs, and solve the problem of high construction safety risk. Moreover, based on the DT evolution model, the construction control platform is developed to accurately output the execution instructions and efficiently guide the whole process of demolition. Through comparison, it is found that the construction accuracy is above 95 %, and the construction resources are efficiently dispatched in a visual form to ensure the construction efficiency.

In this study, an optimization method for cable structure demolition is proposed based on DT. Based on the FE analysis, the visualization model, simulation model and data model are established. On the basis of multi-model fusion, the mapping relationship between construction scheme and mechanical properties is formed, and the efficient optimization of the scheme is realized. The data model based on the improved PSO-BP neural network solves the problems of poor convergence and non-continuity analysis of finite element analysis. On the basis of obtaining the best construction scheme, the tooling configuration, action position and specific process parameters of each construction step are further clarified by relying on the visual model. The defects of FE analysis in construction technology disclosure are solved by visual model. The DT platform is innovatively developed by integrating multi-class models, which improves the efficiency of information sharing and effectively realizes the efficient and accurate guidance of construction.

Based on the proposed method and experimental verification, the virtual-real dynamic mapping and construction scheme optimization can be realized under the drive of DT. For practical engineering, a virtual model can be established according to the structural design file. In the virtual model, the basic information such as the spatial location and material properties of the components is clarified. In the twin simulation module of the DT platform, the mechanical properties corresponding to each construction step are analyzed. Taking the data of the twin simulation module as a sample, the scheme is optimized in the optimization control module. Finally, the construction scheme corresponding to the best change of mechanical properties is obtained. In the demolition modeling module, the construction position of each construction step corresponding to the best scheme and its specific process parameters (such as the application or relaxation of prestress in each construction step) are further clarified. Based on the digital twin platform, the sharing efficiency of construction information is improved, and the efficient

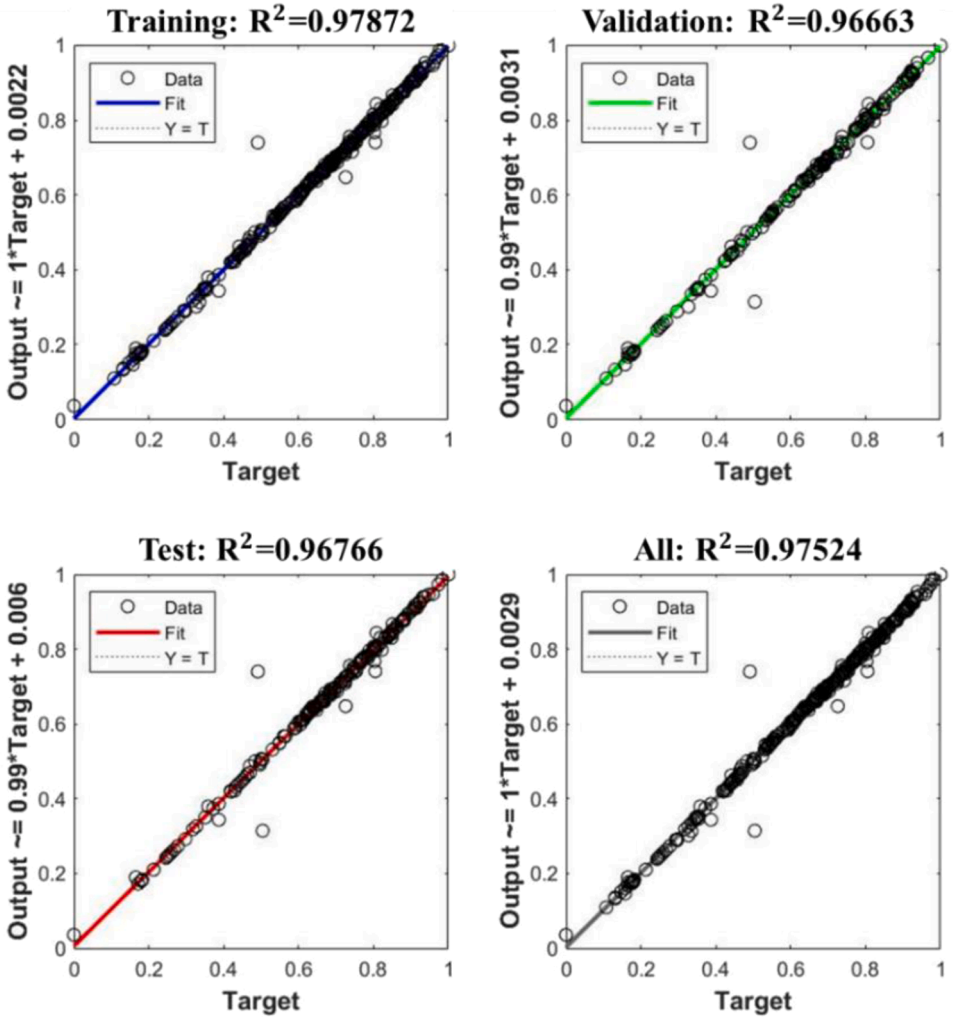
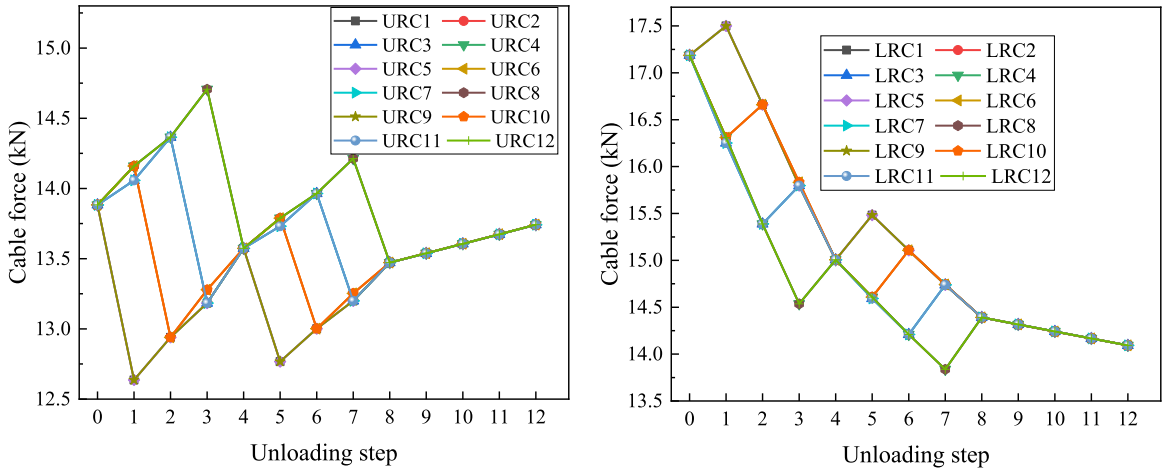


Fig. 12. The analysis accuracy of the sample in the data model.



(a) The variation of URC cable force during unloading process

(b) The variation of LRC cable force during unloading process

Fig. 13. The development of cable force in the process of removing counterweight block.

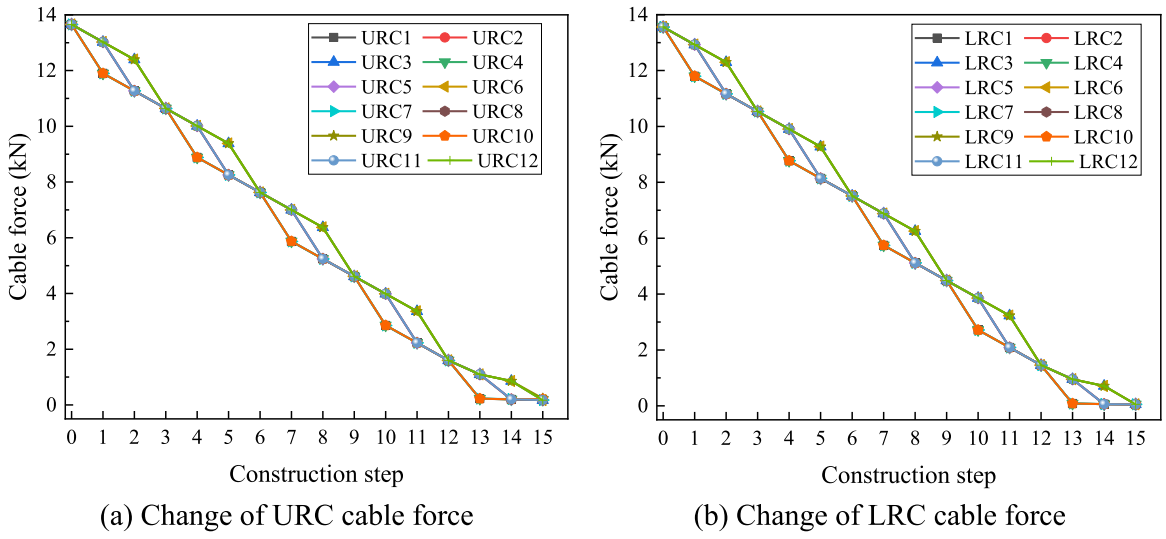


Fig. 14. The development of cable force in the most reasonable relaxation process.

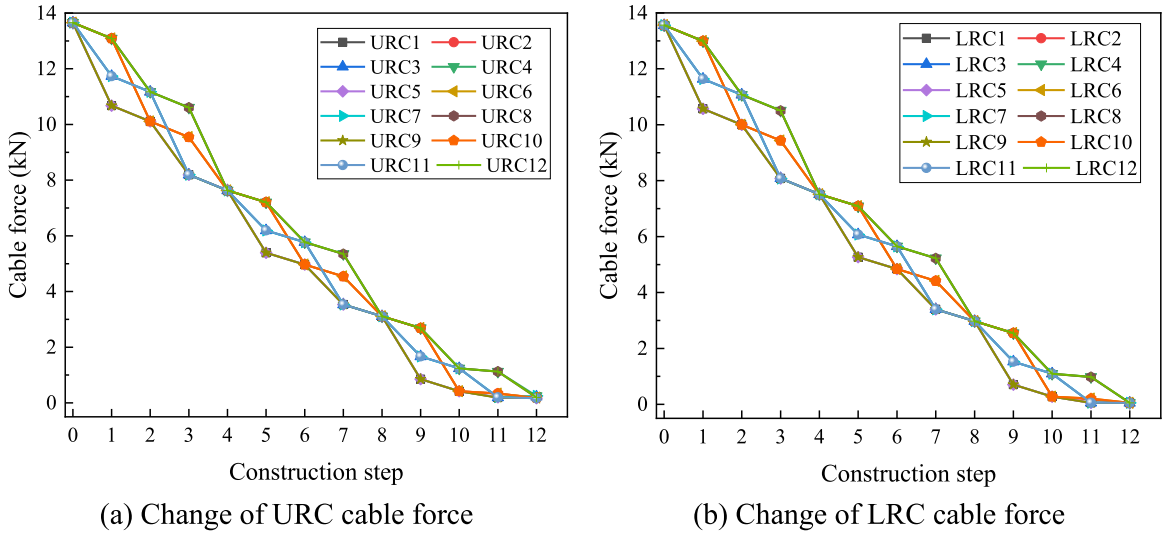


Fig. 15. The development of cable force in the most convenient relaxation process.

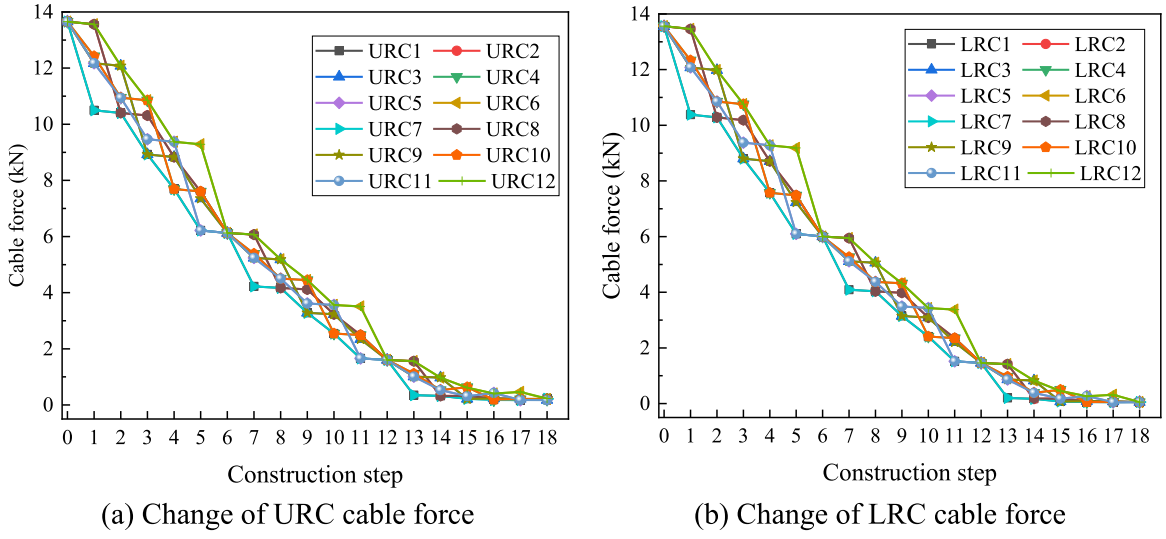


Fig. 16. The development of cable force in the most cost-saving relaxation process.

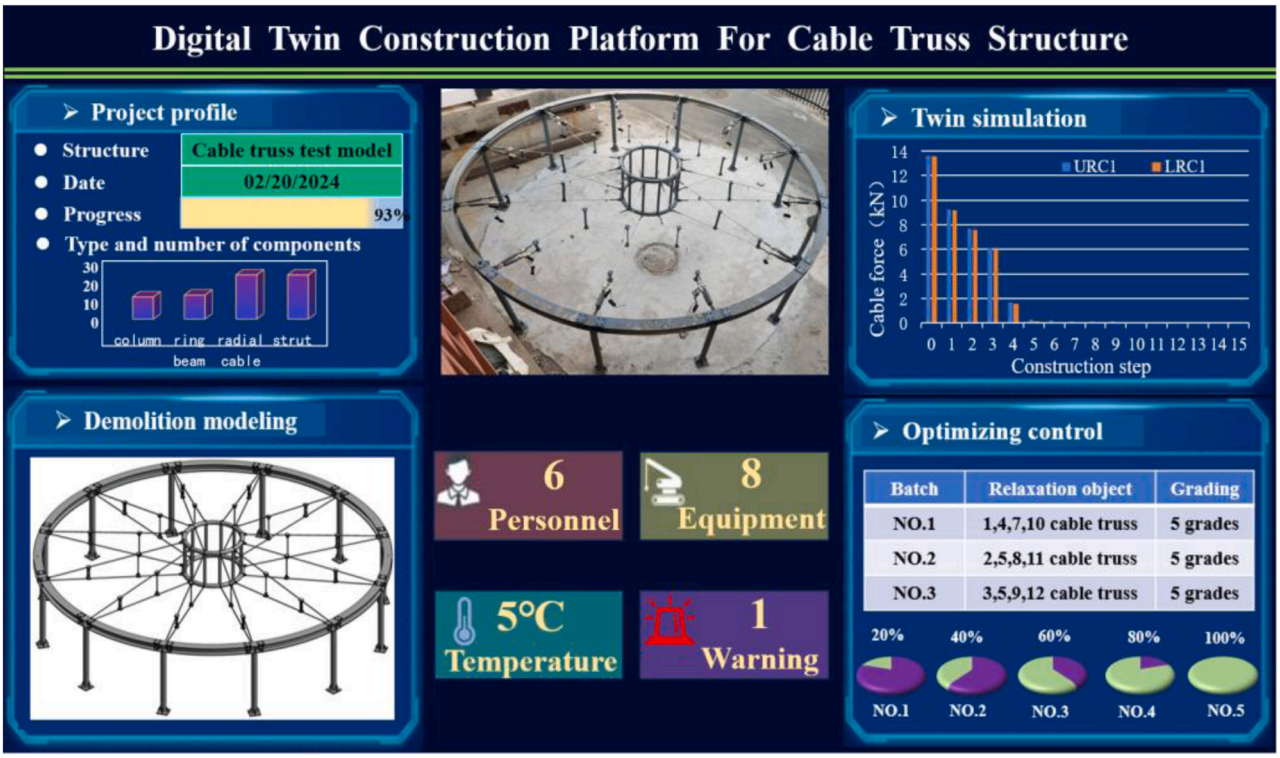


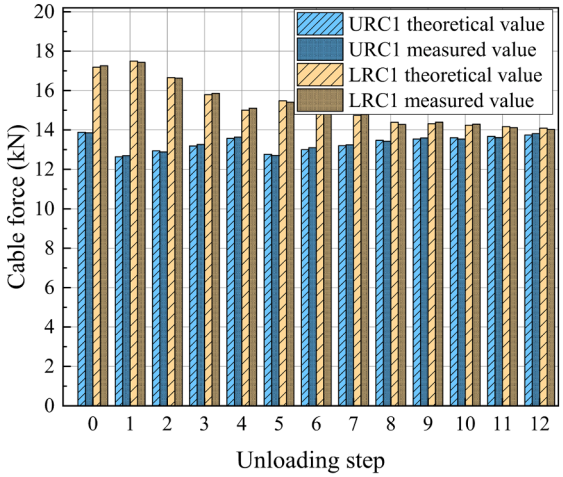
Fig. 17. DT construction control platform.



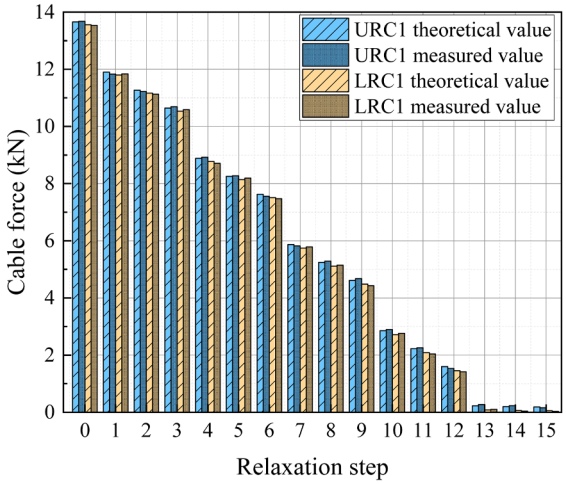
(a) Demolition process of counterweight block



(b) Cable relaxation process



(c) Comparison of MPs of counterweight block demolition



(d) Comparison of cable relaxation MPs

Fig. 18. Demolition process and MPs comparison.

guidance of the demolition process is realized.

In this study, the construction scheme is optimized on the basis of analyzing the MPs of the structure. The construction sequence in the demolition process is analyzed emphatically. The proposed method is verified and applied in the cable truss structure. In addition, the relationship between MPs and construction schemes needs to be further analyzed for other cable structures such as cable domes and truss string structures. For the construction process of large-scale building structures, the multi-functional interface of DT control platform will be fully developed in combination with intelligent equipment such as building robots to realize the construction mode in intelligent construction scenarios.

7. Conclusion

The efficiency of knowledge acquisition in the construction process is relatively low, and it is limited to the experience and knowledge of managers and technicians. In order to solve the problem of high safety risk and low efficiency caused by the cumbersome demolition process and difficult coordination of cable structures, this study proposes a cable structure demolition optimization method driven by the DT evolution model. This study systematically analyzed the evolution process of the demolition DT, and formed a data mapping method between the demolition scheme and the MPs. The best construction scheme is obtained by improving PSO-BPNN. Relying on the DT platform to guide the demolition process in a visual form, it innovatively solves the problems of construction scheduling difficulties, low computational efficiency low information sharing, high security risks and lack of intelligent assistance. In the course of the study, the following conclusions are formed:

(1) From the perspective of virtual-real mapping, the DT evolution process of cable structure demolition is clarified. The virtualization of physical demolition process is realized by integrating visual modeling, simulation modeling and data modeling. Taking full account of the temporal and spatial evolution of the demolition process, a DT evolution model of the demolition process is established. The mapping of physical entities is realized from time, space and application dimensions. Aiming at the demolition execution process, the information flow, component flow, control flow and their relationship are analyzed, and the closed-loop control is realized.
(2) The safety risk is characterized by the MPs of the structure, and the data mapping method between the demolition scheme and the MPs is proposed. The expression language of data association relationship is established, which provides reliable data samples for the optimization of demolition scheme. The PSO-BPNN is improved, and an intelligent analysis method of MPs in the whole process of demolition is proposed. The best solution is obtained by integrating the changes of MPs between each demolition step. The information of the demolition evolution process is divided into functional modules, and the functional framework of the DT platform is built to efficiently guide the actual construction.
(3) The effectiveness of the proposed theoretical method is verified in the cable truss experimental model. In the process of demolition, the best demolition scheme is obtained, which effectively ensures the safety of the structure in the construction process. The DT platform is developed, which integrates the visual model, simulation model and data model of the demolition process, realizes the visual guidance of the demolition scheme, and ensures the efficient scheduling of construction resources and the accuracy of construction.

Under the requirements of adapting to the complex and changing construction environment, this research method realizes the precise optimization of the construction scheme and the efficient control of the construction operation. On the basis of this study, increasing the functional modules of the DT platform and accurately controlling the construction risk under the action of uncertain factors such as the environment are the next research focus.

CRediT authorship contribution statement

Qingwen Zhang: Writing – review & editing, Supervision. **Dechun Lu:** Writing – review & editing, Data curation. **Zhansheng Liu:** Writing – review & editing, Visualization, Project administration, Funding acquisition. **Guoliang Shi:** Writing – original draft, Software, Methodology, Investigation, Formal analysis, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

The authors gratefully acknowledge the financial support provided by Open project of National Civil Engineering Structure Prefabrication and Assembly Engineering Technology Research Center (No. 2023CPCCE-K01) and the National Natural Science Foundation of China (No. 52178095).

References

[1] Vaezzadeh A, Dolatshahi KM. Progressive collapse resistance of cable net structures. J Constr Steel Res 2020;195:107347.
[2] Zurru M. Non-linear normal modes of plane cable trusses. Comput Struct 2021;257:106662.
[3] Asghari R, Abedi K, Chenaghlou MR, Shekastehband B. Retrofit strategies for mitigating progressive collapse of hybrid and traditional cable domes. J Constr Steel Res 2023;204:107807.
[4] Le LX, Siringoringo DM, Katsuchi H, Fujino Y, Luong BX. Stereovision-based vibration measurement of stay cable using synchronized multi-camera setup and video motion magnification. Eng Struct 2023;296:116938.
[5] Chen D, Zhang Y, Qian H, Wang H, Jin X. Numerical approach for simulating the tensioning process of complex prestressed cable-net structures. J Civ Eng Manag 2021;27(8):571–8.
[6] Lu J, Xue SD, Li XY, Dezhkam M. Segmented assembly construction forming method without brackets of spatial cable-truss structure without inner ring cable. Adv Steel Constr 2022;18(3):687–98.
[7] Wolf M, Teizer J, Wolf B, Solberg Buekrue S. A. Investigating hazard recognition in augmented virtuality for personalized feedback in construction safety education and training. Adv Eng Inform 2022;51:101469.
[8] Yao Y, Yan M, Bao Y. Measurement of cable forces for automated monitoring of engineering structures using fiber optic sensors: A review. Autom Constr 2021;126(1):103687.
[9] Chen ZH, Yu YJ, Wang XD, Wu XF, Liu HB. Experimental research on bending performance of structural cable. Constr Build Mater 2015;96(10):279–88.
[10] Guo JM, Zhou D. Pretension simulation and experiment of a negative Gaussian curvature cable dome. Eng Struct 2016;127(15):737–47.
[11] Saleh YN, Mourad SA, Ibrahim AM. Topology optimization of vertical shear links in eccentrically braced frames. Structures 2024;66:106821.
[12] Ghabussi A, Marnani JA, Rohanimanesh MS. Improving seismic performance of portal frame structures with steel curved dampers. Structures 2020;24:27–40.
[13] Ghabussi A, Marnani JA, Rohanimanesh MS. Seismic performance assessment of a novel ductile steel braced frame equipped with steel curved damper. Structures 2021;31:87–97.
[14] Ghabussi A, Mortazavi M, Betha R. Seismic performance of a cold-formed and hot-rolled steel wall system equipped with curved steel dampers. Structures 2023;53:296–316.
[15] Cao ZG, Zhang YS, Wang ZC, Zhao L. Multi-objective optimization of assembly free-form cable-stiffened reticulated shell considering the influence of initial states. Structures 2024;62:106198.
[16] Xu B, Dan D, Han F, Zou YQ. Parameter identification of main cables of cable suspension structures based on vibration monitoring of cable: methodology and experimental verification. J Struct Eng 2021;147(4):04021023.
[17] Liu ZS, Shi GL, Liu Y, Sun Z, Wang JC, Zeng B, Tafsirojjaman T. Investigation of mechanical behaviors of spoke-wheel cable structures through experimental and numerical analysis driven by digital-twin. Structures 2024;62:106099.
[18] Kousta S. Artificial intelligence: Machines that reason. Nature 2016;538:471–6.
[19] Bartlett NW, Tolley MT, Overvelde JTB, Weaver JC, Mosadegh B, Bertoldi K, Whitesides GM, Wood RJ. A 3D-printed, functionally graded soft robot powered by combustion. Science 2015;349(6244):161–5.

[20] Linares-Garcia DA, Roofigari-Esfahan N, Pratt K, Jeon M. Voice-Based Intelligent Virtual Agents (VIVA) to Support Construction Worker Productivity. Autom Constr 2022;143:104554.
[21] Broo DG, Bravo-Haro M, Schooling J. Design and implementation of a smart infrastructure digital twin. Autom Constr 2022;136:104171.
[22] Torzoni M, Manzoni A, Mariani S. Structural health monitoring of civil structures: A diagnostic framework powered by deep metric learning. Comput Struct 2022;271 (15):106858.
[23] Liu JP, Li S, Xu CR, Wu Z, Ao N, Chen YF. Automatic and optimal rebar layout in reinforced concrete structure by decomposed optimization algorithms. Autom Constr 2021;126:103655.
[24] Liu JP, Liu PK, Feng L, Wu WB, Li DS, Chen YF. Automated clash resolution for reinforcement steel design in concrete frames via Q-learning and Building Information Modeling. Autom Constr 2020;112:103062.
[25] Lu JY, Zhang HY, Wu XL. Experimental study on collapse behaviour of truss string structures under cable rupture. J Constr Steel Res 2021;185:106864.
[26] Kamma RC, Jha KN. Quantifying Building Construction and Demolition Waste Using Permit Data. J Constr Eng Manag 2022;148(9):04022091.
[27] Yang JQ, Wu Y, Zhou GY, Xin GY. The dismantling method of wheel-spoke cable-strut tension structures based on experimental and numerical study. Structures 2023;48:1949–63.
[28] Liu ZS, Li H, Liu Y, Wang JC, Tafsirojjaman T, Shi GL. A novel numerical approach and experimental study to evaluate the effect of component failure on spoke-wheel cable structure. J Build Eng 2022;61:105268.
[29] Zhao XZ, Xu ZY, Yan S. Collapse-resisting mechanism and damage propagation pattern of suspended-domes following sudden cable loss. J Constr Steel Res 2024; 213:108338.
[30] Liu WH, Zeng B, Zhou Z, Zheng YF. Theoretical study on progressive collapse of truss string structures under cable rupture. J Constr Steel Res 2022;199:107609.
[31] Chen ZM, Li JY, Zhang SM, Yue QR, Huang ST. Progressive collapse demolition of double-layer space truss under different demolition load conditions. Eng Struct 2023;294:116753.
[32] Technical specification for cable structures: JGJ 257–2012[S]. Beijing: China Architecture & Building Press, 2012. (in Chinese).
[33] Wu K, Soto BD, Zhang F. Spatio-temporal planning for tower cranes in construction projects with simulated annealing. Autom Constr 2020;111:103060.
[34] El-Abbasya MS, Elazounib A, Zayed T. MOSCOPEA: Multi-objective construction scheduling optimization using elitist non-dominated sorting genetic algorithm. Autom Constr 2016;71:153–70.
[35] Shen KC, Li XD, Cao XY, Zhang ZH. Prefabricated construction process optimization based on rework risk. J Constr Eng Manag 2022;148(6):04022031.
[36] Tomczak M, Jaskowski P. Harmonizing construction processes in repetitive construction projects with multiple buildings. Autom Constr 2022;139:104266.
[37] Ma C, Zhao L, Mei XS, Shi H, Yang J. Thermal error compensation of high-speed spindle system based on a modified BP neural network. Int J Adv Manuf Technol 2017;89:3071–85.
[38] Adu-Amankwa NAN, Rahimian FP, Dawood N, Park C. Digital Twins and Blockchain technologies for building lifecycle management. Autom Constr 2023; 155:105064.
[39] Tuhaise VV, Tah JHM, Abanda FH. Technologies for digital twin applications in construction. Autom Constr 2023;152:104931.
[40] Jiang Y. Intelligent building construction management based on BIM digital twin. Comput Intell Neurosci 2021;2021:4979249.
[41] Ritto TG, Rochinha FA. Digital twin, physics-based model, and machine learning applied to damage detection in structures. Mech Syst Signal Process 2021;155(16): 107614.
[42] Li MC, Lu QR, Bai S, Zhang MX, Tian HJ, Qin L. Digital twin-driven virtual sensor approach for safe construction operations of trailing suction hopper dredger. Autom Constr 2021;132:103961.
[43] Teizer J, Johansen KW, Schultz C. The concept of digital twin for construction safety. Constr Res Congr 2022;121:1156–65.
[44] Wang X, Liang CJ, Menassa CC, Kamat VR. Interactive and Immersive Process-Level Digital Twin for Collaborative Human–Robot Construction Work. J Comput Civ Eng 2021;35(6):04021023.
[45] Zhang JY, Cheng JCP, Chen WW, Chen KY. Digital Twins for Construction Sites: Concepts, LoD Definition, and Applications. J Manag Eng 2021;38(2):04021094.
[46] Zhang HY, Lu JY, Gong PF, Li N. Mechanical properties and shape-control abilities of a cable dome under asymmetrical loads. J Constr Steel Res 2023;208:108021.
[47] Xue Y, Luo YZ, Wang YF, Xu X, Wan HP, Shen YB, Fu WW. A new configuration of Geiger-type cable domes with sliding ridge cables: Computational framework and structural feasibility investigation. Eng Struct 2023;286:116028.
[48] Eslamlou AD, Ghaderiaram A, Schlangen E, Fotouhi M. A review on non-destructive evaluation of construction materials and structures using magnetic sensors. Constr Build Mater 2023;397:132460.
[49] Zhang AL, Liu TY, Zhang YX, Liu XC, Jiang ZQ. Experimental and numerical research on mechanical behavior of prestressed high-strength bolt joint. J Constr Steel Res 2021;182:106690.
[50] Salgotra R, Singh U, Singh G, Mittal N, Gandomi AH. A self-adaptive hybridized differential evolution naked mole-rat algorithm for engineering optimization problems. Comput Methods Appl Mech Eng 2021;383:113916.
[51] Naderpour H, Parsa P, Mirrashid M. An Innovative Approach for Moment Capacity Estimation of Spirally Reinforced Concrete Columns Using Swarm Intelligence-Based Algorithms and Neural Network. Pract Period Struct Des Constr 2021;26(4): 04021043.
[52] Zhang Y, Aslani F, Lehane B. Compressive strength of rubberized concrete: Regression and GA-BPNN approaches using ultrasonic pulse velocity. Constr Build Mater 2021;307:124951.
[53] Wang C, Yang QX, Qi TY, Ren WX. A combined method for vehicle load identification based on machine vision and BP neural network. J Civ Struct Health Monit 2023;13:1061–75.
[54] Liu ZS, Shi GL, Liu Y, Sun Z, Wang JC, Zeng B, Tafsirojjaman T. Investigation of mechanical behaviors of spoke-wheel cable structures through experimental and numerical analysis driven by digital-twin. Structures 2024;62:106099.
[55] Shi GL, Liu ZS, Lu DC, Zhang QW, Dezhkam M, Wang ZQ. Design parameter optimization method for a prestressed steel structure driven by multi-factor coupling. Front Struct Civ Eng 2024;18:1066–83.
[56] Shi GL, Liu ZS, Lu DC, Zhang QW, Wang ZQ, Zhao YF. Digital twin - based model updating method for mechanical behaviors analysis of cable truss structure. J Constr Steel Res 2024;221:108917.
[57] Xu J, Xu LJ, Ma Q, Han QH. Force evaluation of internal cable of prestressed grids based on field monitoring and hierarchical objective model updating. J Civ Struct Health Monit 2023;13(2-3):709–27.
[58] Load code for the design of building structures: GB 50009–2012, [S]. Beijing: China Architecture & Building Press, 2012. (in Chinese).
[59] Nguyen QT, Vu VT, Livaoglu R. Artificial Neural Network-based robust technique for period prediction of Ottoman minarets in Türkiye. Structures 2024;61:106087.
[60] Ding MM, Luo B, Guan DZ, Wei Y, Ding SY, Huang LF. Accumulative traction-hoisting construction technology of a semi-rigid steel batten cable dome. Structures 2021;31:159–71.